

Glucose Regulation of Gene Transcription*

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Nutrient gene regulation is an important adaptation allowing survival on intermittent food supplies. This adaptative process exists in all species from yeast to mammals. Glucose, the most abundant monosaccharide in nature, provides a very good example of how organisms have developed regulatory mechanisms to cope with a fluctuating level of nutrient supply. In yeast, glucose facilitates its own use by inducing expression of genes involved in its metabolism while repressing that of those involved in the utilization of alternative carbon sources (for review, see Ref. 1). The mechanisms by which glucose affects gene expression in yeast are now relatively well understood.

In mammals the response to dietary glucose is more complex because it combines effects related to glucose metabolism itself and effects secondary to glucose-dependent hormonal modifications, mainly pancreatic stimulation of insulin secretion and inhibition of glucagon secretion. In the pancreatic β cells, glucose is the primary physiological stimulus for the regulation of insulin synthesis and secretion. In the liver, glucose, in the presence of insulin, induces expression of genes encoding glucose transporters and glycolytic and lipogenic enzymes, e.g. L-type pyruvate kinase (L-PK),¹ acetyl-CoA carboxylase (ACC), and fatty acid synthase, and represses genes of the gluconeogenic pathway, such as the phosphoenolpyruvate carboxykinase gene (for review, see Refs. 2–4). Although insulin and glucagon were long known as critical in regulating gene expression, it is only recently that carbohydrates also have been shown to play a key role in transcriptional regulation (3–5). DNA sequences and DNA binding complexes involved in the glucose-regulated gene expression have been characterized recently in liver and β cells.

The Glucose Signaling Pathway

In most glucose-sensitive tissues, glucose entry is mediated through specific glucose transporters; these include GLUT2, in the liver and β cells, and GLUT4, an insulin-sensitive transporter, in adipocytes and muscle (6). It seems likely that the main role of GLUT2 in gluconeogenic tissues, such as the liver, is to allow for a rapid equilibrium between intra- and extracellular glucose, in particular an easy secretion of glucose under gluconeogenic conditions. In hepatoma cells devoid of GLUT2 and cultured without glucose, the concentration of intracellular glucose 6-phosphate remains high, thus explaining the continuous stimulation of glucose-sensitive genes and consequent loss of glucose responsiveness (7). Accordingly, in GLUT2 $-/-$ mice the intracellular glucose 6-phos-

phate concentration is high in fasting animals (8, 9); and in patients with mutation in the GLUT2 gene (the Fanconi-Bickel syndrome), there is an associated accumulation of intrahepatic glycogen (10). Transcription of the gene for L-PK is also abnormally stimulated in the liver of fasted GLUT2 $-/-$ mice. Transfer of a GLUT2 transgene reestablishes a normal inhibition of glucose utilization by fasting and stimulation by carbohydrate feeding (9).

Guillemin *et al.* (11) have recently suggested that the large intracytoplasmic loop of GLUT2 could also play the role of a signaling molecule. Indeed, hyperexpression of this loop in mhAT3F hepatoma cells (7) abrogates glucose responsiveness.

After its entry into liver, adipocytes, and β cells, glucose has to be metabolized to generate an intracellular signal that allows for transcriptional regulation of metabolic genes. Although it is now clear that the first step of glycolysis, namely phosphorylation of glucose to glucose 6-phosphate, is instrumental for glucose-dependent regulation (either positive or negative), the subsequent steps are still disputed and the exact nature of the active intermediate remains obscure (3, 5). In the liver, glucose 6-phosphate can be used in glycolysis, the pentose phosphate pathway, glycogen synthesis, and hexosamine synthesis. Work from our laboratory using xylitol, an intermediate of the pentose phosphate pathway, has suggested that in hepatocytes in primary culture the glucose signal could be generated through the non-oxidative branch of the pentose phosphate pathway (12). This result has been further extended *in vivo* by Massillon *et al.* (13), who noted that xylitol mimics the effect of hyperglycemia on glucose-regulated genes without a change in intracellular glucose 6-phosphate concentration in the liver. Alternatively, there are several reports suggesting that glucose 6-phosphate itself is a signaling molecule (3, 14). According to Mourrillas *et al.* (15), only production of glucose 6-phosphate correlates accurately with the induction of glucose-regulated genes. Finally, for a specific subset of genes regulated by glucose through its metabolism in muscle and fat, the hexosamine biosynthetic pathway was reported to mediate the transcriptional effects of glucose on gene expression (16, 17).

Kinase Cascade and Role of AMPK

It is likely that a phosphorylation/dephosphorylation cycle is involved in signaling the availability of glucose to the transcriptional machinery. In this regard, protein phosphatase inhibitors prevent the effects of glucose on L-PK gene expression² and other glucose-regulated genes (18–20). The involvement of the cAMP-dependent protein kinase in regulating the glucose-dependent signaling pathway is well documented (4, 21, 22). More recently, another protein kinase, the AMP-activated protein kinase (AMPK), has been suggested to play a key role in transmission of the glucose signal for transcription. AMPK is a serine/threonine kinase acting as a key metabolic “master switch” by phosphorylating target proteins involved in carbohydrate and fat metabolism in response to changes in cellular energy charge. AMPK is activated by stresses that deplete ATP, leading to a rise in the AMP/ATP ratio within the cell (23). AMPK is a heterotrimeric complex of a catalytic (α) and two regulatory subunit (β and γ) proteins. Proteins related to all three subunits have been characterized in the yeast Snf1p complex (24). There are two catalytic subunit isoforms encoded by two different genes, namely $\alpha 1$ and $\alpha 2$ (23). Interestingly, although $\alpha 1$ -containing complexes are exclusively cytoplasmic, $\alpha 2$ complexes are found both in the nucleus and cytoplasm (25, 26).

AMPK activity was first reported to be decreased in β cell lines incubated in elevated glucose concentrations (26, 27). We and others have shown that the AMPK activator, 5-amino-4-imidazolecarboxamide riboside, inhibits the glucose-dependent activation of several genes in hepatocytes in primary culture (19, 28). Thus, inactivation of AMPK may restore the transcriptional activity of

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¹ The abbreviations used are: L-PK, L-type pyruvate kinase; ACC, acetyl-CoA carboxylase; S14, spot 14; AMPK, AMP-activated protein kinase; GIRE, glucose response element; ChoRE, carbohydrate response element; COUP-TF, chicken ovalbumin upstream promoter-transcription factor; USF, upstream stimulatory factor; SREBP, sterol response element-binding protein; PI3K, phosphatidylinositol 3-kinase; PDX1, pancreatic duodenum homeobox 1.

² B. Doiron, unpublished data.

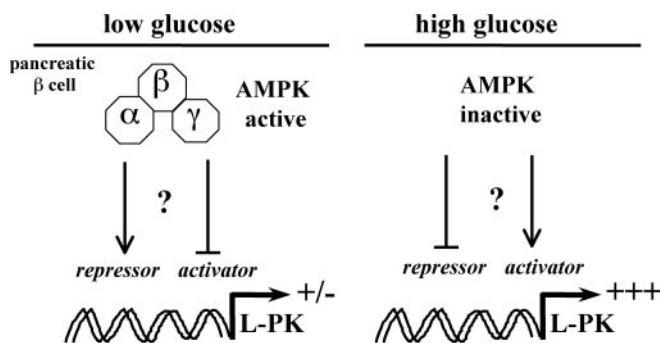


FIG. 1. Role of AMPK in the regulation by glucose of the L-PK gene transcription in pancreatic β cell. When cultured under low glucose conditions, AMPK activity is elevated and L-PK gene transcription is low. AMPK activity is inhibited by an elevated glucose concentration leading to an increase in L-PK gene transcription. Microinjection of antibodies to the $\alpha 2$ subunit of AMPK mimics the effects of elevated glucose on the L-PK promoter activity (26).

glucose-responsive genes. To test this hypothesis, the group of Rutter in collaboration with our laboratory (26) used the single-cell antibody microinjection strategy, coupled with dynamic imaging of luciferase reporter constructs. These studies demonstrated that microinjection of antibodies against $\alpha 2$, but not the $\alpha 1$ AMPK catalytic subunit, mimics the effects of elevated glucose on the L-PK and preproinsulin promoters in MIN6 β cells. Interestingly, in each case the effects were only observed when antibody injection was performed both in the nucleus and the cytosol, indicating the importance of either a cytosolic phosphorylation event and/or the subcellular localization of the $\alpha 2$ subunits. Incubation with 5-amino-4-imidazolecarboxamide riboside diminished, but did not abolish, the effect of glucose on preproinsulin transcription (26). These data suggest that glucose-induced changes in AMPK activity are necessary and sufficient for the regulation of the L-PK gene by this sugar (see Fig. 1) and that they play an important role in the regulation of the preproinsulin gene promoter. This result is of interest with regard to what is known about yeast. In *Saccharomyces cerevisiae*, the Snf1 kinase complex is activated by glucose removal, which results in phosphorylation-dependent inactivation of transcription factors involved in gene repression, such as Mig1 (1, 24). In mammalian cells, AMPK seems to have the opposite effect on gene expression; instead of inhibiting a repressor, AMPK inhibits an activator or activates a repressor (Fig. 1). Possible targets of AMPK are currently being investigated in our laboratory, and the generation of AMPK knock-out mice is in progress.

The Glucose Response Elements of L-PK and S14 Genes

To assess the effect of glucose on gene transcription, our laboratory has extensively studied the regulation of the rat L-PK gene expression by glucose. We defined a DNA sequence responsible for mediating the positive response to glucose *ex vivo* and *in vivo* (29–32). This sequence, termed the glucose response element (GIRE), is closely related to the carbohydrate response element (ChoRE) described by the group of Towle (33, 34) in the regulatory region of the S14 gene. A similar glucose-activatable region has been recently identified in the promoter of the gene for the glucagon receptor (35). The complex that binds to the GIRE/ChoRE of the L-PK and S14 genes cooperates with an adjacent DNA binding site, termed the auxiliary site, to confer strong glucose responsiveness (29, 31, 34). However, oligomerized GIRE/ChoRE sequences placed in front of a heterologous promoter are sufficient, in absence of the auxiliary site, to confer the carbohydrate response (29, 31, 34).

The L-PK GIRE consists of two palindromic non-canonical E boxes (CANNTG) separated by 5 base pairs. These E boxes bind the upstream stimulatory factors (USFs), which are members of the basic helix-loop-helix leucine zipper family (31, 34). We have shown that in the liver, as well as in most tissues tested, USF binding activity is mainly accounted for by the USF1/USF2 heterodimer (36). Our results using cells and gene knock-out mice indicate that the endogenous USFs are important for a kinetically normal activation of various diet-dependent genes by glucose (32, 37–39). However, these factors cannot, by themselves, explain the transcriptional regulation

of glucose-responsive genes by glucose. Indeed, most genes whose promoters include USF-binding E boxes are not regulated by glucose. In addition, the glucose responsiveness conferred by GIREs/ChoREs is not parallel to their affinity for USFs (40). Finally, USF synthesis as well as DNA binding activity do not appear to be regulated by the diet. We therefore sought novel partners of the glucose response complex using the one-hybrid system in yeast. We identified, in addition to USF, chicken ovalbumin upstream promoter-transcription factor II (COUP-TFII), an orphan nuclear receptor of the steroid/thyroid hormone receptor superfamily. COUP-TFII binds to the GIRE *in vitro*, and COUP-TF-containing complexes that interact with the GIRE are present in liver nuclear extracts (41). The COUP-TF and USF binding sites are overlapping, and binding of one factor interferes with binding of the other. Consequently, overexpression of COUP-TFII inhibits USF-dependent transactivation of the L-PK gene promoter and also represses its stimulation by glucose in hepatocytes in primary culture. Furthermore, a mutated GIRE binding site with very low affinity for COUP-TFII impaired the glucose response because of increased activity under low glucose conditions (41). We propose that COUP-TFII-containing dimers are involved in a glucose sensor system abrogating transactivation by USFs in the absence of glucose. This sensor complex could be regulated by interaction with other partner(s) sensitive to glucose, that we are currently looking for.

Other Putative Partners of the GIRE/ChoRE of L-PK and S14

There are several recent reports concerning GIRE DNA-binding proteins. Hasegawa *et al.* (42) reported that carbohydrate refeeding resulted in an increase in the concentration of a novel L-PK GIRE binding activity in the liver. They excluded USF, Sp1, c-Myc, and HNF4 as being involved in this activity. So far, the cognate DNA-binding proteins have not been purified. Yamada *et al.* (43) described the purification of two novel GIRE-binding proteins that may be related to COUP-TFII. Finally, Koo and Towle (44) have recently characterized a new ChoRE binding activity that correlates with glucose-dependent transcriptional activity. This DNA binding activity is only recovered with oligonucleotides capable of conferring a glucose response and not with mutants of the sites that are inactive in supporting the glucose response. This binding activity does not seem to be regulated by dietary glucose. The authors proposed a new model for the ChoRE consisting of two E box half-sites related to CACG motifs.

Recently, the sterol response element-binding protein 1c (SREBP-1c) (45, 46) was also proposed as a mediator of glucose induction of genes encoding proteins of glycolytic and lipogenic pathways, including L-PK and S14 genes (47). However, this effect of SREBP-1c on glucose-responsive genes is likely to reflect the role of SREBP-1c as a mediator of insulin action (as discussed in detail in the third minireview of this series by Osborne (81)) rather than its direct involvement in the glucose signaling pathway. Indeed, the L-PK GIRE has a very low affinity for SREBP (44), which is, in addition, a very poor transactivator of the L-PK gene promoter (48, 49). Accordingly, complete SREBP-1 deficiency in knock-out mice results only in a partial abrogation of L-PK gene expression in response to carbohydrate refeeding, whereas insulin-responsive genes such as glucose-6-phosphate dehydrogenase and glycerol 3-phosphate acyltransferase are totally unresponsive (50). In fact, the impaired glucose response of the L-PK gene in the liver of SREBP1 $-/-$ mice could result from the deficiency in glucokinase, the gene transcription of which is insulin/SREBP-1c-dependent (51). Indeed, glucokinase induction by insulin/SREBP-1c is needed for glucose phosphorylation to glucose 6-phosphate, an indispensable intermediate of glucose-dependent gene activation.

A summary of the role of glucose and insulin in the transcriptional regulation of glycolytic and lipogenic genes in the liver is presented in Fig. 2.

Transcriptional Factors of the Sp1 Family Are Involved in Glucose Responsiveness

Several reports have suggested that the ubiquitously expressed transcription factor Sp1 may also provide a mechanism for glucose responsiveness. This was first reported for the ACC gene in adipocytes where glucose was able to induce an increase of Sp1 binding activity (52). This induction was not due to an increase in the

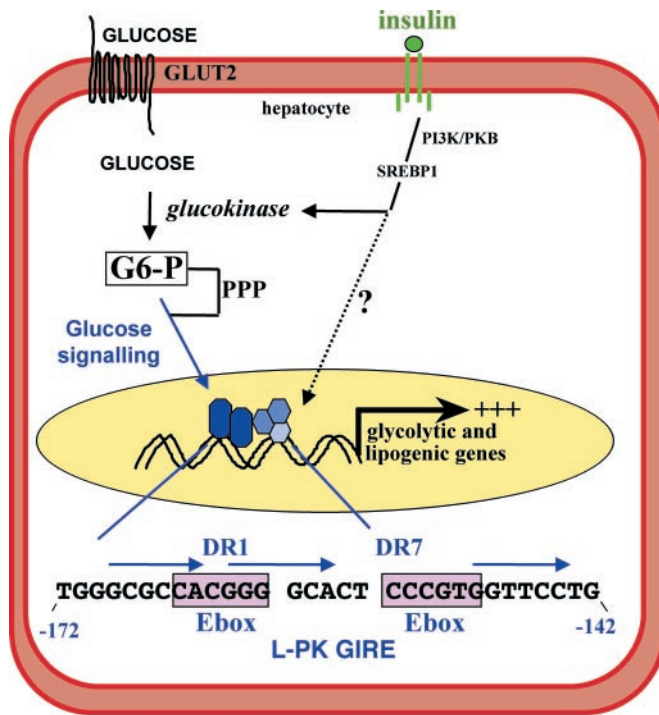


FIG. 2. Role of glucose and insulin in the transcriptional regulation of glycolytic and lipogenic genes in the liver. The involvement of the PI3K/protein kinase B (PKB) pathway in the insulin-dependent activation of the glucokinase gene transcription has been demonstrated recently (80). An overview of the functional DNA element (–144 to –172 base pairs from the start site of transcription of the rat L-PK gene) involved in the glucose-dependent regulation of the L-PK gene transcription is presented. The E boxes and direct repeat (DR1 and DR7) elements in the GIRE are indicated. PPP, pentose phosphate pathway.

amount of Sp1 but to dephosphorylation of the existing Sp1 in the nucleus. Glucose treatment increases the amount of protein phosphatase 1 catalytic subunit resulting in the activation of Sp1 binding activity on the ACC promoter (18). A glucose-dependent Sp1 dephosphorylation resulting in a higher DNA binding activity has been confirmed (53). Members of the Sp1 family have also been proposed to be required for glucose-dependent induction of the plasminogen activator inhibitor 1 gene (54) and the transforming growth factor β (16) gene through the glucosamine pathway in vascular muscle cells. Finally, factors of the Sp1 family were reported to be somehow involved in regulation of glucose/insulin-dependent genes encoding leptin, fatty acid synthase, and ATP citrate-lyase (55, 56). In the case of ATP citrate-lyase, the level of expression of Sp1 and Sp3 factors was reported to be directly regulated by glucose (57). It is interesting to note that in the absence of adequate nutrition, Sp1 becomes hypoglycosylated and thereby subject to proteolytic degradation (58).

Transcriptional Regulation of Insulin Gene Expression in Pancreatic β Cells: Role of PDX1

An impressive breakthrough in the field of gene regulation by glucose involves the control of the insulin gene promoter by the transcription factor pancreatic duodenum homeobox protein, PDX1 (IPF-1/STF-1/IUF-1/IDX-1), in pancreatic β cells. PDX1 was first described as a key factor for lineage determination of the developing endocrine pancreas (59, 60). Strong evidence now supports a role for this homeodomain transcription factor in the mechanism whereby glucose stimulates insulin gene transcription in β cells. Increased glucose concentration in the mature islet leads to PDX1 phosphorylation and subsequent translocation into the nucleus, where it binds to the insulin gene promoter resulting in its transcriptional activation (61, 62). Glucose-dependent activation of PDX1 nuclear translocation occurs through a β cell signaling pathway involving PI3K. Today, involvement of p38/SAPK2 in the downstream pathway is disputed (62, 63). Other metabolizable nutrients that induce insulin release (including fructose, pyruvate, and xylitol), as well as insulin itself, were also shown to activate PDX1 DNA binding activity and insulin promoter activity through

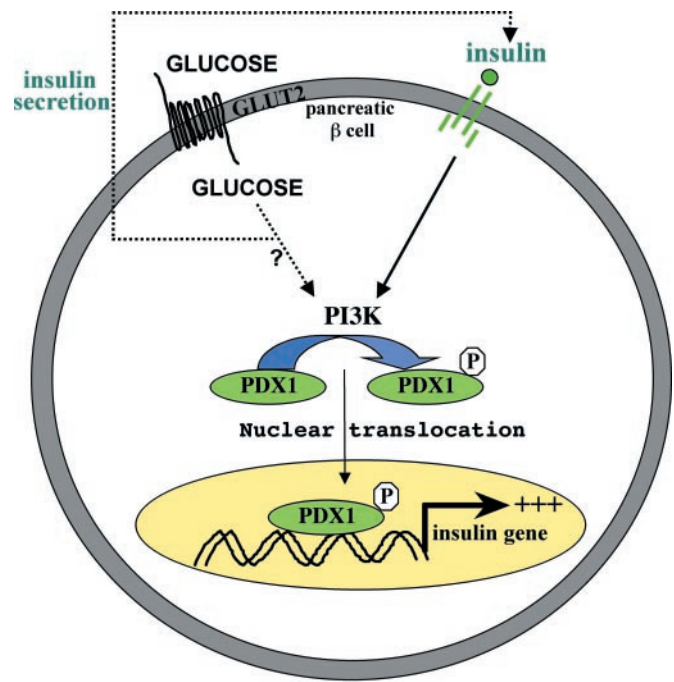


FIG. 3. Role of glucose and insulin in the stimulation of PDX1 DNA binding activity and insulin gene transcription in the pancreatic β cell. Both insulin and glucose were shown to activate insulin gene transcription via a pathway involving PI3K and stress-activated protein kinase 2 (SAPK2) (64). However, whether glucose is able to activate PI3K independently of stimulating insulin secretion has not been determined.

PI3K (64). Further experiments are now needed to determine whether the effects of insulin and glucose are additive or whether glucose acts as an insulin secretagogue, whereas PDX1 and proinsulin gene activation are triggered by insulin (65). A schematic view of glucose and insulin regulation of PDX1 DNA binding and insulin gene promoter activity is presented in Fig. 3.

The pivotal role of PDX1 in the nutrient regulation of insulin mRNA production was further assessed using pancreatic β cell lines. These studies showed that impaired glucose-regulated insulin secretion and insulin content are associated with dysregulated PDX1 expression (66, 67). The role of PDX1 in diabetes has been increasingly documented over the past several years. A dominant negative mutation in PDX1 was shown to cause maturity-onset diabetes of the young; MODY4 is an autosomal dominant form of type 2 diabetes characterized by an early onset and a primary defect of insulin secretion (68, 69). Furthermore, new mutations in PDX1 that may predispose to type 2 diabetes in humans were reported, their phenotype depending upon the nature of the mutation (70, 71). Very recently, Ferber *et al.* (72, 73) have demonstrated that *in vivo* transduction of hepatocytes with a recombinant adenovirus-PDX1 vector reactivates the silent proinsulin gene, as well as genes for proconvertases PC1/PC3; this reinforces the view that PDX1 is a major transactivator of β cell-specific genes.

Other glucose-dependent genes, such as the L-PK (26, 74), are also regulated in β cells, and the mechanisms underlying their regulation are under investigation. Recently, Josefsen *et al.* (75) using a subtraction cloning strategy isolated the zinc finger transcription factor Egr-1 and demonstrated that its expression is up-regulated in primary islet cells in response to glucose. In view of the close homology of Egr-1 with transcription factors that regulate carbohydrate metabolism in yeast, the authors propose that Egr-1 might play a pivotal role in response to sustained glucose stimulation. The genes regulated by Egr-1 in β cells are currently unknown.

Conclusions

There are clearly different mechanisms of gene activation in mammalian tissues induced by dietary carbohydrate. Some genes are mainly activated by insulin, whose secretion is increased by glucose; this is the case for SREBP1 and glucokinase in the liver (Fig. 2) and may also be true for the proinsulin gene in β cells. Hepatic SREBP1 and PDX1 in β cells appear to be major insulin-

dependent transactivators of gene transcription. Some genes are also inhibited by insulin and the role of different winged helix transcription factors or forkhead; a family of transcription factors such as FKHR has been well documented recently (76–78). The insulin signaling pathways, mediating either positive or negative effects on the transcriptional machinery, mainly pass through PI3K activation (79, 80).

Other genes are regulated by glucose itself in the presence of glucose-phosphorylating enzymes, namely glucokinase, where the role of AMPK has been demonstrated in both the liver and β cells. However, the transcription factors involved in the glucose response differ according to the regulated genes and currently remain elusive. In fact, it is probable that there is no single regulator of gene expression by glucose but rather that glucose-dependent cues allow the cross-talk between factors and most likely regulatory co-factors to determine the transcriptional activation in a cell-specific manner. The specificity of the response could be achieved through post-translational modifications, mainly phosphorylation of various tissue-specific or ubiquitous factors. Nuclear translocation, as exemplified for PDX1 in β cells and Mig1 in yeast, offers an additional level of regulation. Alternatively, the concentration of transcription factors could be directly regulated by glucose. We predict that progress in deciphering the mechanisms of glucose-dependent gene regulation will quickly follow those made in the last several years for insulin control of gene transcription.

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